

High Dimensional Post Selection OLS under Multicollinearity

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Abstract

High-dimensional settings, where the number of covariates (p) far exceeds the number of observations (n), present significant challenges for classical Ordinary Least Squares (OLS) estimation. The presence of multicollinearity in these settings poses even further challenges. This dissertation conducts a comprehensive simulation study to compare the performance of three variable selection methods in such settings: Lasso, Threshold Ridge (TR) tuned via Leave-One-Out Cross-Validation (TR-LOOCV), and a novel TR variant tuned using the Bayesian Information Criterion (TR-BIC). Models are evaluated on four key metrics: prediction risk, true support recovery, coefficient estimation error, and the empirical coverage of confidence intervals. Our findings demonstrate that TR-BIC achieves superior support recovery and more reliable coverage, particularly in the presence of extreme multicollinearity. A key component of this dissertation is an investigation into the suitability of the TR-BIC method for causal inference within a Post-Double-Selection OLS framework.

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1 Introduction

1.1 The Challenge of High-Dimensionality and Multicollinearity

High-dimensional settings, where the number of potential covariates (p) is large relative to the sample size (n), are becoming increasingly common in empirical economics. This setting can occur organically, when datasets are inherently high-dimensional, e.g administrative data, text data etc. It can also be created artificially when researchers construct a larger set of variables from transformations and interactions of a smaller set of variables, to allow for more flexible functional forms.

In these $p \gg n$ settings, classical Ordinary Least Squares (OLS) fails due to its tendency to severely overfit the data, leading to poor out-of-sample performance and unreliable parameter estimates (Belloni et al, 2014). To deal with such settings, regularization may be used to impose a 'lower-dimensional structure', typically by assuming that only a small subset of covariates are truly relevant - the *approximate sparsity assumption*.

The workhorse estimator in these settings is the Least Absolute Shrinkage and Selection Operator (LASSO) (Tibshirani, 1996). The LASSO estimator works by minimizing the sum of squared residuals subject to a constraint on the sum of the absolute values of the coefficients (the l_1 -norm). The geometry of this l_1 -norm constraint is what gives the LASSO its variable selection property; its shape allows the optimisation process to set some lesser important coefficient estimates to exactly zero, effectively removing them from the model.

However, a common feature of modern economic datasets is multicollinearity. For example, in financial econometrics, various measures of the yield curve (e.g., curve spreads) or different measures of inflation (e.g., CPI or PCE etc.) often exhibit multicollinearity. Under these conditions, the performance of the LASSO estimator can be unsatisfactory. Because the l_1 -penalty is indifferent between equally predictive but highly correlated variables, the selector may become unstable; small changes in the data can cause it to arbitrarily select one variable from a correlated group while discarding the others (Zou & Hsiao, 2005). This can make finding stable and easily interpretable models of underlying economic relationships challenging.

This limitation motivates the investigation of alternative methods that are more robust to multicollinearity. One such alternative is the Threshold Ridge (TR) estimator (Shao & Deng 2012), a two-step procedure that first uses ridge regression (l_2 penalty) to manage multicollinearity and then applies a hard threshold to achieve sparsity. Standard tuning of the regularisation constant and threshold is done via leave-one-out cross-validation (LOOCV) optimised for predictive accuracy. However, in many economic applications, the goal is not prediction but consistent model identification for structural or

causal analysis. For this purpose, a criterion that prioritises model selection consistency and simplicity may be more appropriate. This motivates the central methodological contribution of this dissertation: the introduction of a novel TR variant tuned using the Bayesian Information Criterion (**TR-BIC**). The BIC selects a model by balancing goodness-of-fit with model complexity, imposing a penalty on the number of included parameters that is known to favour more parsimonious models (Schwarz, 1978). We hypothesise that by imposing this stricter penalty, TR-BIC will achieve more accurate support recovery, particularly in the presence of high multicollinearity.

1.2 Post-Selection OLS and the Inference Problem

A common workflow in applied research is to first use a method like LASSO to perform variable selection and then apply OLS to this subset, a procedure known as Post-Selection OLS (PS-OLS). This refitting step is intended to correct the shrinkage bias introduced by the regularization. This two-step process, however, renders standard statistical inference invalid. Because the model itself is selected using the same data, the final model is not fixed, but is in fact a random outcome of the selection procedure (Leeb & Pötscher, 2005). Standard inferential theory ignores this additional source of randomness, which results in distorted sampling distributions and confidence intervals that fail to achieve their nominal coverage rates (Lee et al., 2016).

Formal methods exist to address this post-selection inference (PSI) problem. For obtaining valid inference on a *specific* target parameter, such as a causal effect, the 'double-selection' procedure is a particularly robust framework. It constructs estimating equations that are orthogonal to, or immune to, small errors in the selection of nuisance (control) variables. It delivers valid confidence intervals for the target parameter even if the variable selection for the controls is imperfect (Belloni et al 2014). This dissertation leverages this framework in an empirical application to investigate whether our proposed TR-BIC selector can improve the performance of the double-selection procedure when faced with multicollinearity.

1.3 Research Questions

This dissertation conducts a detailed investigation of PS-OLS estimators, focusing on the challenges of multicollinearity and the performance of the proposed TR-BIC selector. The central research questions are as follows:

- How do different first-stage selectors; specifically LASSO, TR-LOOCV, and our proposed TR-BIC, affect the performance of PS-OLS estimators under varying degrees of multicollinearity?

- Does the TR-BIC selector, designed for model selection consistency, provide superior support recovery and more reliable subsequent inference compared to existing methods, particularly under **extreme** multicollinearity?
- How do simulation results compare to existing theoretical risk bounds for post-selection estimators?
- In an empirical application, can the TR-BIC selector, when used within a double-selection framework, provide reliable estimates and confidence intervals for a target causal parameter compared to the standard LASSO-based approach?

1.4 Structure of paper

The remainder of this dissertation is structured as follows. Section 2 reviews the relevant literature on high-dimensional regression, variable selection, and post-selection inference. Section 3 outlines the theoretical framework and the benchmark estimators. Section 4 details the simulation methodology, including the data generating processes and evaluation metrics. Section 5 presents and discusses the simulation results. Section 6 applies the TR-BIC method to an empirical study of a key causal question in economics. Finally, Section 7 concludes with a summary of findings and directions for future research.

2 Literature Review

High-dimensional settings present a fundamental challenge to classical econometric methods. In $p \gg n$ settings, OLS is no longer viable as the Gram matrix $X^\top X$ becomes singular, meaning the OLS estimator is not uniquely identifiable. Even when $p \leq n$, a large number of predictors can cause OLS to overfit the data, leading to poor out-of-sample performance and unreliable inference.

To make the analysis tractable, it is common to impose a lower-dimensional structure on the problem (Belloni et al., 2012). The main structural assumption in the literature is that of *approximate sparsity*. Within the context of the linear model, $y = X\beta^* + \epsilon$, this assumption posits that the true coefficient vector β^* is sparse, meaning most of its components are zero or near-zero. Formally, the size of the true support, $s^* = \|\beta^*\|_0$, is assumed to be much smaller than the number of candidate covariates, i.e., $s \ll p$. The core challenge, therefore, becomes the accurate identification of this true, sparse support.

2.1 Penalized Regression and Variable Selection

A range of penalized regression techniques have been developed to discover the true support s^* . The most widely used of these is the Least Absolute Shrinkage and Selection Operator (LASSO), introduced by Tibshirani (1996), which solves the following optimisation problem:

$$\hat{\beta}_{\text{LASSO}} \in \arg \min_{\beta \in \mathbb{R}^p} \left(\frac{1}{n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right)$$

The ℓ_1 -penalty term, $\|\beta\|_1$, induces sparsity by shrinking the coefficients of less important variables to be exactly zero. The regularization parameter, λ , governs the degree of shrinkage and is critical to the estimator's performance. In practice, λ is often chosen via cross-validation (CV) to optimise out-of-sample predictive performance (Chetverikov et al., 2021). An alternative, theory-grounded approach is the Rigorous LASSO (rLasso) of Belloni et al. (2012), which employs a data-driven penalty designed to be robust to the heteroskedasticity common in economic data.

Despite its widespread use, the LASSO has well-documented limitations, most notably its instability in the presence of multicollinearity.

2.2 The Challenge of Multicollinearity

Multicollinearity, a condition where columns of the design matrix X are highly correlated, also leads to a near-singular Gram matrix $X^\top X$. This severely degrades the performance of many variable selectors by inflating the variance of coefficient estimates. For the LASSO, this instability manifests in its tendency to arbitrarily select only one variable from a group of highly correlated variables, while setting the coefficients of the others to

zero. This choice can be highly sensitive to small perturbations in the data, making the selected model unstable and difficult to interpret (Zou & Hastie, 2005).

Several methods have been proposed to address this tendency. The Elastic Net (Zou & Hastie, 2005) adds an ℓ_2 -penalty to the LASSO objective function, which encourages a 'grouping effect' where the coefficients of correlated variables are shrunk towards each other. While the Elastic Net combines the ℓ_1 and ℓ_2 penalties into a single convex optimisation problem, the Threshold Ridge (TR) estimator by Shao & Deng (2012) offers a different approach by explicitly separating the tasks of stabilisation and selection into two distinct stages.

The TR estimator first computes a Ridge Regression estimator to stabilise the coefficients in the presence of multicollinearity:

$$\hat{\beta}_{\text{Ridge}} = (X^\top X + h_n I_p)^{-1} X^\top y$$

Second, a hard-thresholding step is applied to the components of $\hat{\beta}_{\text{Ridge}}$ to induce sparsity and select the final model. By first stabilising estimates before enforcing sparsity, the TR estimator is designed to avoid the arbitrary selection problem inherent in the LASSO.

From a theoretical standpoint, the TR estimator holds a distinct advantage in settings with high collinearity. The prediction risk of the Post-Lasso estimator has an upper bound of the order $O(s \log(p)/n)$ (Belloni & Chernozhukov, 2013). The $\log(p)$ term indicates that the risk necessarily grows with the total number of candidate predictors. In contrast, the prediction risk for a Threshold Ridge estimator has been shown to be of the order $O(s \log(\kappa)/n)$, where κ is the effective rank of the design matrix $X^\top X$ (Shao & Deng, 2012; Zhang & Politis, 2023). In the presence of high collinearity, the effective rank κ can be much smaller than p , making the TR bound strictly tighter. A central goal of this dissertation is to systematically evaluate these theoretical properties via simulation.

Correct tuning of the TR estimator's two parameters—the ridge penalty h_n and the thresholding value a_n —is crucial. While the original proposal by Shao & Deng (2012) employs a form of leave-one-out cross-validation (LOOCV) to optimise predictive accuracy, many economic applications prioritise model parsimony. The Bayesian Information Criterion (BIC) is a well-established method to balance model fit and complexity. Standard BIC is defined as $\text{BIC} = n \log(\text{RSS}/n) + k \log(n)$, where k is the number of estimated parameters. For high-dimensional settings, this is often extended (EBIC) to include a penalty on the size of the model space, $\text{EBIC} = n \log(\text{RSS}/n) + k \log(n) + 2\gamma k \log(p)$ (Chen & Chen, 2008). Tuning the TR estimator via such information criteria remains largely unexplored in the literature, representing a key research gap this dissertation aims to address.

2.3 Post-Selection Estimation and Inference

In applied work, variable selection is often the first step in a two-stage procedure. After a selector identifies a sparse model with support $\hat{S}$, a second 'refitting' step is performed.

2.3.1 Post-Selection OLS (PS-OLS) Estimators

A common refitting method is Post-Selection OLS (PS-OLS), where OLS is run on the sub-matrix of selected covariates, $X_{\hat{S}}$, to obtain the final coefficient estimates: $\hat{\beta}_{\text{PS-OLS}} = (X_{\hat{S}}^{\top} X_{\hat{S}})^{-1} X_{\hat{S}}^{\top} y$. The primary motivation for this step is to mitigate the shrinkage bias inherent in penalized estimators like the LASSO, which systematically shrink non-zero coefficients towards zero (Belloni & Chernozhukov, 2013; Hansen et al., 2014).

The theoretical properties of PS-OLS are well-understood. Belloni & Chernozhukov (2013) demonstrate that the Post-Lasso estimator can achieve a near-oracle rate of convergence for the estimation error, performing at least as well as the LASSO. Crucially, their framework, which applies to any 'good generic selector', provides the theoretical motivation for this dissertation's investigation into the performance of OLS following selection by Threshold Ridge.

2.3.2 The Challenge of Post-Selection Inference (PSI)

While PS-OLS is an effective estimation strategy, it introduces a fundamental problem for statistical inference. As highlighted by Leeb & Pötscher (2005), standard OLS confidence intervals and p-values are generally invalid following a data-driven model selection step. The selection process is a random event dependent on the data, and ignoring this randomness effectively truncates the sampling distribution of the final estimators. This causes naïve confidence intervals to have coverage rates well below their nominal levels, making standard inference unreliable (Lee et al., 2016; Zhang et al., 2022).

While a substantial body of recent literature has been devoted to developing valid Post-Selection Inference (PSI) methods, much of this work has centered on the LASSO. Prominent approaches include the *debiased LASSO*, which constructs an asymptotically normal estimator, $\sqrt{n}(\hat{\beta}_{\text{DB}} - \beta^*)$ (e.g., van de Geer et al., 2014; Zhang & Zhang, 2014), and *exact conditional inference*, which derives the exact post-selection distribution by conditioning on the selection event $\{\hat{M}(y) = M\}$ (Lee et al., 2016). For inference on a specific causal parameter, the *double selection* or *double machine learning* framework offers a robust solution built on Neyman Orthogonality (Belloni, Chernozhukov, & Hansen, 2014; Chernozhukov et al., 2018). However, an investigation of practical and valid inferential methods following selection by alternatives like the Threshold Ridge estimator remains a less developed area, further motivating this dissertation's empirical investigation.

2.4 Summary and Contributions

This dissertation contributes to the literature on high-dimensional variable selection and post-selection inference, with a particular focus on the challenges posed by multicollinearity. We have identified several key areas where further investigation is warranted. This dissertation aims to fill these gaps through the following principal contributions:

First, it introduces and evaluates a novel tuning method for the Threshold Ridge estimator: the **Threshold Ridge with Bayesian Information Criterion (TR-BIC)**. Building on the work of Shao and Deng (2012), we assess whether TR-BIC yields more parsimonious and accurate support recovery than the original TR-LOOCV, especially under severe multicollinearity. Second, we conduct a comprehensive simulation study to systematically assess the impact of multicollinearity on the finite-sample performance of Post-Selection OLS estimators. This provides an empirical test of the theoretical risk bounds that suggest Threshold Ridge is more robust to multicollinearity than the LASSO. Third, we evaluate the empirical coverage of confidence intervals for refitted OLS coefficients, offering a practical measure of inferential reliability following model selection. Finally, we apply the TR-BIC selector within the double-selection framework to a canonical dataset in development economics, demonstrating its practical utility.

3 Theoretical Framework and Estimators

In this section, we introduce our theoretical framework. We begin by formally specifying the high-dimensional linear model and the approximate sparsity assumption. We then define the estimators that will be compared throughout this study. The Oracle estimator, which provides a theoretical performance lower bound; the Lasso and Elastic Net; and the Threshold Ridge estimator. We also introduce the novel Threshold Ridge variant tuned via the Bayesian Information Criterion. Finally, we review the existing theoretical risk bounds for these methods.

3.1 Model Specification

We consider the standard high-dimensional linear regression model:

$$y = X\beta^* + \epsilon$$

where $y \in \mathbb{R}^n$ is the vector of observed outcomes, $X \in \mathbb{R}^{n \times p}$ is the design matrix of p covariates for n observations, $\beta^* \in \mathbb{R}^p$ is the vector of true, unknown coefficients, and $\epsilon \in \mathbb{R}^n$ is a vector of unobserved random errors. We operate in the high-dimensional setting where the number of covariates is much larger than the sample size, $p \gg n$. The error terms are assumed to be independent and identically distributed (i.i.d.) following a normal distribution, $\epsilon_i \sim N(0, \sigma^2)$, with $\sigma^2 = 1$ without loss of generality.

To make estimation and inference tractable in this high-dimensional setting, we impose a lower-dimensional structure on the problem by assuming that the true coefficient vector β^* is *sparse*. This means that only a small number of covariates are truly relevant for explaining the data generating process of y . Formally, we define the true support (the set of indices of non-zero coefficients) as $S^* = \{j : \beta_j^* \neq 0\}$, and we assume its cardinality, $s = |S^*|$, is small relative to the sample size, $s \ll n$.

A central focus of this dissertation is the impact of multicollinearity. The rows of X , denoted $x_i^\top$, are drawn i.i.d. from a multivariate normal distribution, $x_i \sim N(0, \Sigma)$, where $\Sigma \in \mathbb{R}^{p \times p}$ is the population covariance matrix. To induce multicollinearity, we use a Toeplitz covariance matrix, defined as:

$$\Sigma_{jk} = \rho^{|j-k|} \quad \text{for } \rho \in [0, 1)$$

This structure induces local correlation, where covariates that are closer in index are more highly correlated. The parameter ρ directly controls the degree of multicollinearity. For the simulations, the design matrix X is generated by first drawing $Z \in \mathbb{R}^{n \times p}$ with $Z_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$ and then setting $X = ZL^\top$, where L is the Cholesky factor of Σ (i.e., $\Sigma = LL^\top$).

3.2 The Oracle Estimator: *An Infeasible Benchmark*

To establish a theoretical performance benchmark, we define the Oracle Ordinary Least Squares (OLS) estimator $\hat{\beta}_{\text{oracle}}$. The oracle estimator is constructed by applying OLS only to the subset of covariates corresponding to the true support, X_{S^*} :

$$\hat{\beta}_{\text{oracle}} = (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top y$$

The oracle represents the best possible performance achievable with a linear estimator. Its prediction risk provides a lower bound for any feasible estimation procedure. Under standard assumptions, the oracle risk is given by (Proof in Appendix B):

$$R_{\text{oracle}} = \mathbb{E} \left[\|X(\hat{\beta}_{\text{oracle}} - \beta^*)\|_2^2 \right] = \sigma^2 s$$

This implies a mean squared prediction error (MSPE) of $\frac{\sigma^2 s}{n}$.

3.3 The Lasso Estimator

The Least Absolute Shrinkage and Selection Operator (Lasso), introduced by Tibshirani (1996), is the workhorse estimator for high-dimensional sparse regression models. It is defined as the solution to the following ℓ_1 -penalized least squares problem:

$$\hat{\beta}_{\text{Lasso}}(\lambda) = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

The ℓ_1 -norm penalty, $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$, simultaneously shrinks coefficients towards zero and sets some of them to be exactly zero, thus performing variable selection. The performance of the Lasso is critically dependent on the choice of the tuning parameter, λ . We will consider two principal methods for selecting λ :

Cross-Validation (Lasso-CV): The penalty λ is chosen to minimize the out-of-sample prediction error, estimated via K-fold cross-validation. As shown by Chetverikov et al. (2021), this data-driven choice for λ yields a Lasso estimator with nearly optimal rates of convergence.

Rigorous Lasso (rLasso): Proposed by Belloni et al. (2012), this approach uses a theory-based, data-driven penalty level, given by $\lambda = c \cdot \hat{\sigma} \cdot \Lambda(1 - \gamma|X)$, where $\Lambda(1 - \gamma|X)$ is the $(1 - \gamma)$ -quantile of the normalized score process.

3.4 The Elastic Net Estimator

To address Lasso's instability under high multicollinearity, Zou and Hastie (2005) proposed the Elastic Net. It uses a penalty that is a convex combination of the ℓ_1 and ℓ_2

norms:

$$\hat{\beta}_{\text{ENET}}(\lambda, \alpha) = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right) \right\}$$

The ℓ_2 component encourages a 'grouping effect', where correlated predictors are selected or discarded together, while the ℓ_1 component enforces sparsity. The mixing parameter $\alpha \in [0, 1]$ and the penalty strength λ are tuned via cross-validation.

3.5 The Threshold Ridge (TR) Estimator

The Threshold Ridge (TR) estimator, proposed by Shao and Deng (2012), is a two-step procedure designed to separate the task of stabilisation from that of variable selection.

3.5.1 Step 1: Ridge Regression for Stabilisation

First, a Ridge regression estimator is computed to obtain stabilised, non-sparse coefficient estimates. The Ridge estimator solves the ℓ_2 -penalized problem:

$$\hat{\theta}(h) = \arg \min_{\theta \in \mathbb{R}^p} \{ \|y - X\theta\|_2^2 + h \|\theta\|_2^2 \}$$

where $h > 0$ is the ridge penalty parameter.

Proposition 3.1 (Closed-form Ridge Solution) *The unique minimizer of the Ridge objective function is given by:*

$$\hat{\theta}(h) = (X^\top X + hI_p)^{-1} X^\top y.$$

The objective function is convex and differentiable. Setting the gradient with respect to θ to zero yields:

$$\begin{aligned} \nabla_{\theta} ((y - X\theta)^\top (y - X\theta) + h\theta^\top \theta) &= 0 \\ -2X^\top (y - X\theta) + 2h\theta &= 0 \\ X^\top X\theta - X^\top y + h\theta &= 0 \\ (X^\top X + hI_p)\theta &= X^\top y. \end{aligned}$$

Since $X^\top X$ is positive semi-definite and $h > 0$, the matrix $(X^\top X + hI_p)$ is strictly positive definite and therefore invertible, which gives the unique solution. The term hI_p adds a positive constant to the diagonal of the potentially singular Gram matrix $X^\top X$, ensuring invertibility and providing a unique, stable solution even under severe multicollinearity.

3.5.2 Step 2: Hard-Thresholding for Sparsity

Second, a hard-thresholding operator is applied to the Ridge coefficients to induce sparsity. The final TR estimator is defined component-wise as:

$$\tilde{\beta}_j(h, a_n) = \hat{\theta}_j(h) \cdot \mathbf{1}\{|\hat{\theta}_j(h)| > a_n\}$$

where a_n is the thresholding parameter and $\mathbf{1}\{\cdot\}$ is the indicator function. Only coefficients with a magnitude greater than a_n are retained, with the selected support being $\hat{S}(h, a_n) = \{j : |\tilde{\beta}_j(h, a_n)| \neq 0\}$. Following Shao and Deng (2012), we parameterize the threshold as $a_n = C_1 n^{-\alpha}$ for constants $C_1 > 0$ and $0 < \alpha \leq 1/2$.

3.5.3 Tuning the TR Estimator

The performance of the TR estimator depends on the choice of the two tuning parameters, (h, a_n) . We investigate two tuning strategies:

TR-LOOCV Tuning for Predictive Accuracy

Shao and Deng (2012) propose tuning (h, a_n) by minimizing a computationally efficient proxy for the Leave-One-Out Cross-Validation (LOOCV) error.

Lemma 3.1 (Shao and Deng (2012, Lemma 1)) *For any linear smoother S with $\hat{y} = S y$, leaving out i satisfies*

$$y_i - \hat{y}_i^{\setminus i} = \frac{y_i - \hat{y}_i}{1 - S_{ii}}.$$

Applying this lemma with $S = P(h) = X(X^\top X + hI_p)^{-1}X^\top$ and the same $\tilde{\beta}(h, a_n)$ (nonlinear only through thresholding), one obtains the approximation for LOOCV as:

$$\text{CV}(h, a_n) \approx \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - x_i^\top \tilde{\beta}(h, a_n)}{1 - P_{ii}(h)} \right)^2,$$

as in Shao and Deng (2012, Sec. 2.3). Thus one need only compute the diagonal entries

$$P_{ii}(h) = [X(X^\top X + hI_p)^{-1}X^\top]_{ii}.$$

A grid search is performed over a range of (h, a_n) values to minimize this criterion.

TR-BIC Tuning for Model Selection (Novel Proposal)

We propose a new tuning method where (h, a_n) are chosen to minimize the Bayesian Information Criterion (BIC), which is known to favor more parsimonious models. For high-dimensional settings, we use an extended BIC (Chen & Chen, 2008):

$$\text{EBIC}(h, a_n) = n \ln(\text{RSS}/n) + k \ln(n) + 2\gamma k \ln(p)$$

where $\text{RSS} = \|y - X\tilde{\beta}(h, a_n)\|_2^2$, $k = \|\tilde{\beta}(h, a_n)\|_0$, and $\gamma \in [0, 1]$ is a hyperparameter controlling the penalty on model complexity. This criterion explicitly balances goodness-of-fit with both the number of selected parameters and the size of the original model space. We refer to the EBIC as simply BIC throughout the rest of this paper.

3.6 Existing Theoretical Risk Bounds for PS-OLS

The theoretical properties of Post-Selection OLS (PS-OLS) estimators have been studied extensively. The upper bounds on the prediction risk provide a key motivation for this dissertation's focus:

Post-Lasso & Post-Elastic Net: Belloni & Chernozhukov, 2013 show that the prediction risk for these estimators is bounded by an expression of the order

$$O(\sigma^2 \frac{s \log p}{n})$$

The presence of the $\log p$ term indicates that the risk grows with the total number of candidate predictors.

Post-Threshold Ridge: In contrast, the risk for a refitted TR estimator has been shown to be of the order

$$O(\sigma^2 \frac{s \log \kappa}{n})$$

where $\kappa = \text{tr}((X^\top X + hI_p)^{-1} X^\top X)$ is the effective rank of the design matrix (Shao & Deng, 2012; Zhang & Politis, 2023). In the presence of high multicollinearity, the eigenvalues of $X^\top X$ decay rapidly, and the effective rank κ can be much smaller than p . This provides a strong theoretical motivation for hypothesising that TR-based methods will have a distinct advantage in high-dimensional, multicollinear settings, as their performance depends on the effective number of dimensions rather than the ambient dimension of the covariate space.

4 Methodology: Simulation Design

This section details the design of our simulation. We specify the data generating processes discussed in section 3.1. We then detail the practical implementation of the estimators, and define the metrics used for their evaluation.

4.1 Data Generating Process (DGP)

We generate data according to the sparse linear model $y = X\beta^* + \epsilon$, as specified in Section 3.1. Simulations are repeated 1000 times per parameter combination for completeness. The key parameters underlying the data generating process are summarised in Table 1.

A critical feature of our experimental design is sample splitting. Within each replication, the generated sample of $n = 100$ observations is divided into two equal and independent halves: a *selection set* ($n_{\text{select}} = 50$) used for tuning and training the variable selection models, and an *inference set* ($n_{\text{infer}} = 50$) used exclusively for the second-stage OLS refitting. This procedure ensures that the evaluation of the refitted model is not contaminated by overfitting from the selection stage. Furthermore, an independent test set of $n_{\text{test}} = 200$ observations is drawn from the same DGP to provide an unbiased estimate of out-of-sample prediction risk.

Table 1: Data Generating Process (DGP) Parameters

Parameter	Symbol	Values and Description
Sample Size	n	100 (split into $n_{\text{select}} = 50$ and $n_{\text{infer}} = 50$)
Covariate Dimension	p	$\{250, 500\}$
True Sparsity	$s = S^* $	25
Coefficient Values	β_j^*	1.0 for $j \in S^*$, and 0 otherwise.
Correlation Structure	Σ	Toeplitz matrix: $\Sigma_{kl} = \rho^{ k-l }$.
Correlation Parameter	ρ	$\{0.05, 0.6, 0.9\}$ (Weak, Strong, Severe).
Error Distribution	ϵ	i.i.d. $N(0, \sigma^2)$, with $\sigma^2 = 1$.
Replications	N	1000 per configuration.

4.2 Implementation of Comparative Estimators

The estimators defined in section 3 are implemented using this sample-splitting scheme. As an *infeasible* benchmark, we compute the Oracle OLS estimator by applying OLS to the known true predictors on the *inference* data.

The Post-Lasso (PL) selection stage is performed using 5-fold cross-validation via the `LassoCV` function from the `scikit-learn` library. Similarly, the Post-Elastic Net (PEN) is tuned using `ElasticNetCV` to select both the penalty strength and the ℓ_1 -ratio. We

also include the Post-Rigorous Lasso (PrLasso), which employs the theory-based penalty selection method on the *selection* data.

The core of our investigation compares two implementations of Post-Threshold Ridge (PTR). Both are tuned via a grid search over the ridge penalty h_n and the hard-thresholding value a_n . The PTR-LOOCV estimator selects the h_n and a_n pair that minimizes the Leave-One-Out Cross-Validation error, as detailed in Section 3.5.3. In contrast, our proposed PTR-BIC estimator selects the h_n and a_n pair that minimizes the Bayesian Information Criterion.

4.3 Evaluation Metrics

The performance of each estimator is then evaluated based on the metrics defined in Table 2.

Table 2: Evaluation Metrics

Metric	Formula	Description
Prediction Risk	$\frac{1}{n_{\text{test}}} \ X_{\text{test}}(\hat{\beta} - \beta^*)\ _2^2$	Out-of-sample prediction error.
Estimation Error	$\ \hat{\beta} - \beta^*\ _2^2$	ℓ_2 error of the coefficient vector.
<i>Support Recovery</i>		
True Positive Rate	$\frac{ \hat{S} \cap S^* }{ S^* }$	Fraction of true variables selected.
False Positive Rate	$\frac{ \hat{S} \setminus S^* }{p - S^* }$	Fraction of false variables selected.
Model Size	$ \hat{S} $	Total number of selected variables.
<i>Inference</i>		
Empirical Coverage	$\text{Prop}(\beta_j^* \in \text{CI}_j^{95\%})$	Coverage of naïve 95% CI for a true β_j^* .

These metrics allow us to assess the trade-offs each method makes, particularly under increasing levels of multicollinearity. For instance, a method may achieve low prediction risk, but at the cost of poor true support recovery, which may lead to suboptimal post-selection inference downstream.

4.4 Computational Details

All simulations were conducted in Python, primarily using the `scikit-learn`, `statsmodels`, and `NumPy` libraries. The execution of the 1000 replications for each scenario was parallelized and run on the CSD3 high-performance computing cluster at the University of Cambridge.

5 Simulation Results and Discussion

This section presents and interprets the results of the Monte Carlo simulations detailed in section 4.

5.1 Model Selection

Our aim in high-dimensional settings is the accurate identification of the true, sparse support given a large set of potential covariates. The accuracy of this model selection affects the accuracy of later Inference. We observe large differences between estimators tuned for predictive accuracy versus those tuned for model selection consistency.

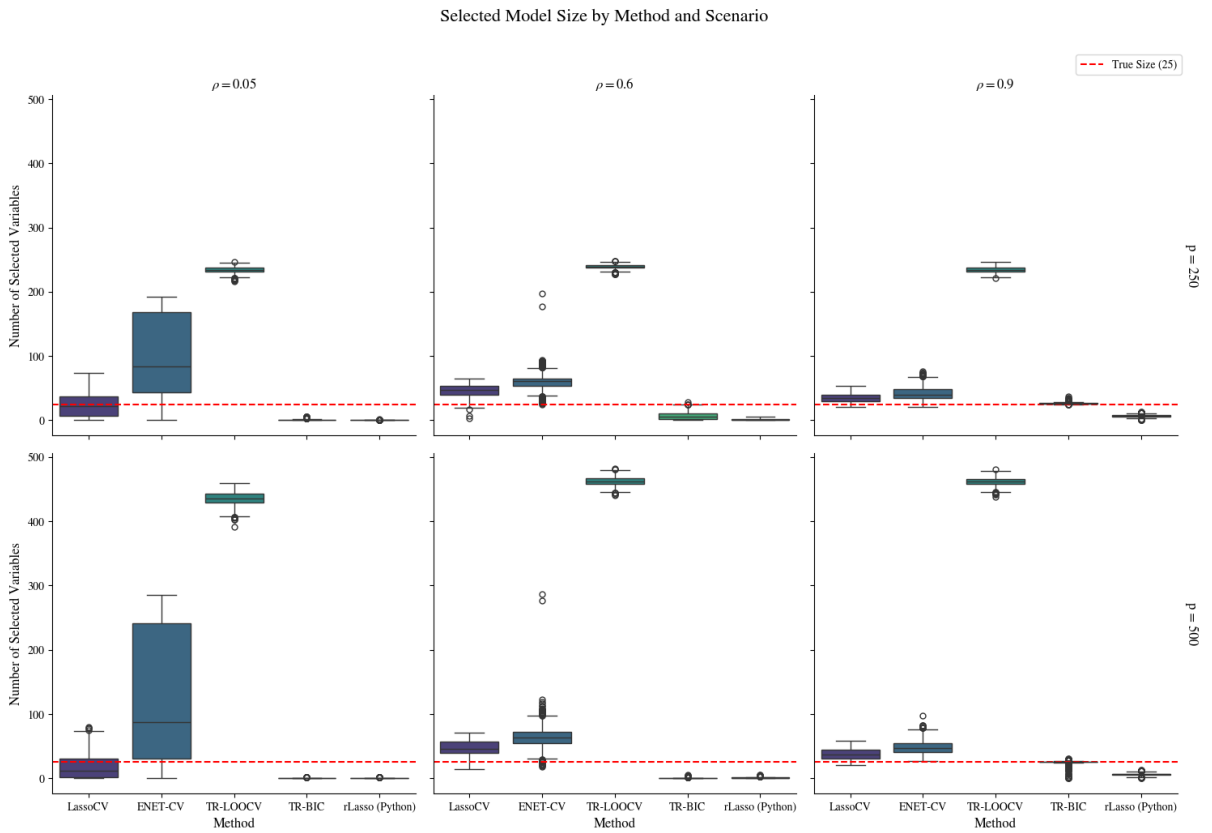


Figure 1: Distribution of Selected Model Size by Method and Scenario. The red dashed line indicates the true model size ($s = 25$)

Figure 1 plots the distribution of selected model sizes by multicollinearity strength.

The TR-LOOCV estimator, which optimises for predictive accuracy, consistently selects a dense model where the number of included covariates approaches the total number of predictors ($|\hat{S}| \approx p$). This may occur because its Leave-One-Out criterion favours a minimal hard threshold if the inclusion of numerous noise variables can reduce its out-of-sample prediction error.

In contrast, our proposed TR-BIC demonstrates mixed performance. It seems that for low ρ settings, the signal from individual true predictors is attenuated by the initial ridge

shrinkage, such that the coefficients become too insignificant to overcome the BIC’s fixed penalty ($k \ln n$), leading to severe under-selection. For high $\rho = 0.9$, however, the ridge step seems to properly regularise the block of correlated true predictors, which allows TR-BIC to obtain a model size close to the true sparsity, s .

The theory-driven rLasso is uniformly too conservative, with its strong penalty resulting in a null model in nearly all scenarios.

The ℓ_1 -penalized methods, LassoCV and ENET-CV, offer a middle ground, consistently over-selecting relative to s but remaining more parsimonious than TR-LOOCV. However the extreme variance of ENET-CV at low correlations ($\rho = 0.05$) may reflect some instability in its mixing hyperparameter, α , across simulations.

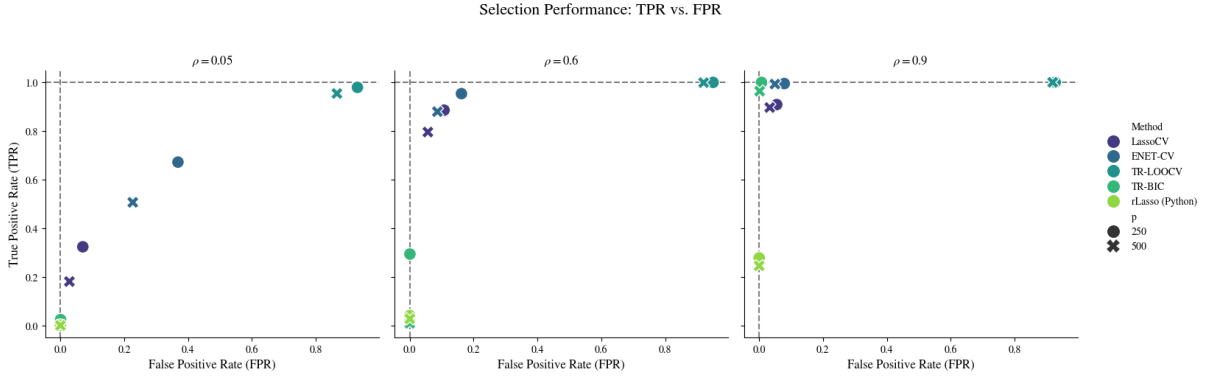


Figure 2: Selection Performance: True Positive Rate (TPR) vs. False Positive Rate (FPR).

Figure 2 plots the True Positive Rate (TPR) against the False Positive Rate (FPR) for each method. An Oracle would achieve performance located at the (0, 1) point). The results show a clear performance difference depending on varying degrees of multicollinearity. In the $\rho = 0.9$ scenario, the TR-BIC estimator performs best, achieving a TPR approaching 1.0 while maintaining an FPR close to zero. In contrast, the TR-LOOCV estimator consistently yields a TPR of 1.0 but at the cost of an extremely high FPR (approximately 0.9), confirming that its optimisation for predictive accuracy does not yield a parsimonious model.

The LassoCV and ENET-CV estimators exhibit robust performance in the high-multicollinearity setting ($\rho = 0.9$), achieving TPRs above 0.9 and FPRs below 0.1. Nevertheless, in this high-correlation regime, the TR-BIC method achieves the best performance.

5.2 Prediction, Estimation, and the Multicollinearity Trade-off

Whilst accurate model selection is important, the utility of an estimator can often lie in its predictive accuracy and the precision of its coefficient estimates. We now discuss the results between the trade-offs inherent in model selection.

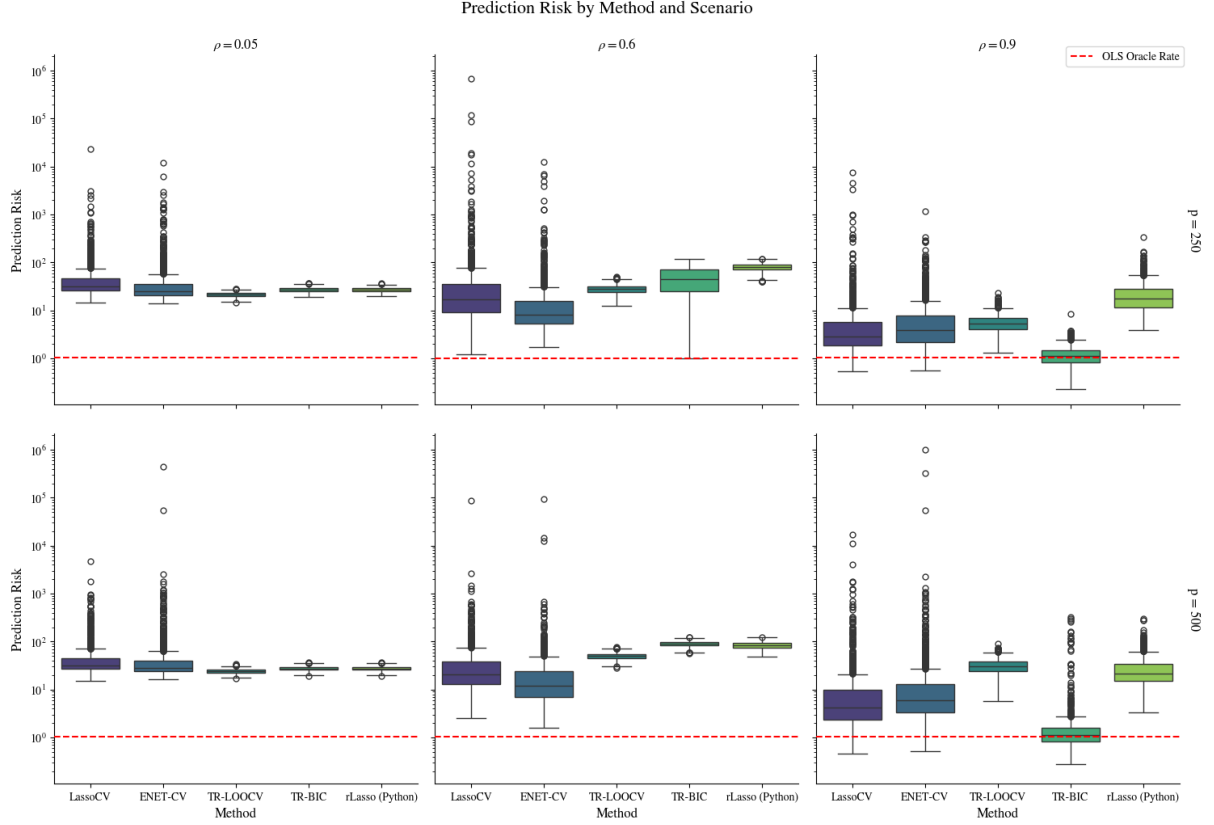


Figure 3: Prediction Risk by Method and Scenario (log scale). The red dashed line indicates the risk of the OLS Oracle estimator. Note TR-BIC’s superior performance at $\rho = 0.9$.

Figure 3 displays the out-of-sample prediction risk on a logarithmic scale. In the presence of severe multicollinearity ($\rho = 0.9$), the superior model selection of TR-BIC translates directly into superior predictive performance. Its median risk is the lowest among all feasible methods and approaches that of the OLS Oracle. However, a reversal occurs at lower levels of correlation. For $\rho \in \{0.05, 0.6\}$, the dense models selected by TR-LOOCV are predictively competitive with, and sometimes superior to, the parsimonious models from TR-BIC. This can be interpreted as a form of stabilisation through inclusion; by averaging over a large set of weakly correlated predictors during OLS refitting, the estimator achieves a lower variance, albeit at the cost of interpretability and coefficient bias.

The LassoCV and ElasticNetCV estimators exhibit strong predictive performance across all simulation settings, but with large outliers. This may be due to some instability of the ℓ_1 selection procedure when tuned via cross-validation. The set of selected covariates may be highly variable across replications, leading to occasional poor model selections that produce these high-risk outliers.

The quality of the coefficient estimates (essential for valid inference), is examined in Figure 4. The sampling distribution of a single true coefficient estimate reveals that

TR-BIC is unique in its ability to deliver a nearly unbiased estimate under severe multicollinearity, with its distribution centred on the true value of 1.0 when $\rho = 0.9$. Both Lasso and ENET also demonstrate strong performance, while the TR-LOOCV estimates are visibly attenuated towards zero. The rLasso performs poorly in settings characterised by high multicollinearity ($\rho \geq 0.6$).

In the low-collinearity settings, however, the TR-BIC fails to select variables in the true support, thus disallowing coefficient estimation entirely. This may occur because, as mentioned above, the TR-BIC regularization may be too strong and shrinking all coefficients to near zero. The BIC penalty may then remove these shrunken coefficients, as their inclusion may not warrant the additional cost to the BIC.

We now consider the overall ℓ_2 estimation error, $\|\hat{\beta} - \beta^*\|_2^2$, as reported in the summary table 4. Counter-intuitively, the heavily biased TR-LOOCV estimator often exhibits the lowest total estimation error. This may occur because the ℓ_2 norm aggregates errors across all p coefficients. While this may be optimal for minimizing a specific loss function, it is undesirable when focused on the structural interpretation or causal effect of a specific covariate. For such purposes, the lower bias offered by TR-BIC is a more valuable property.

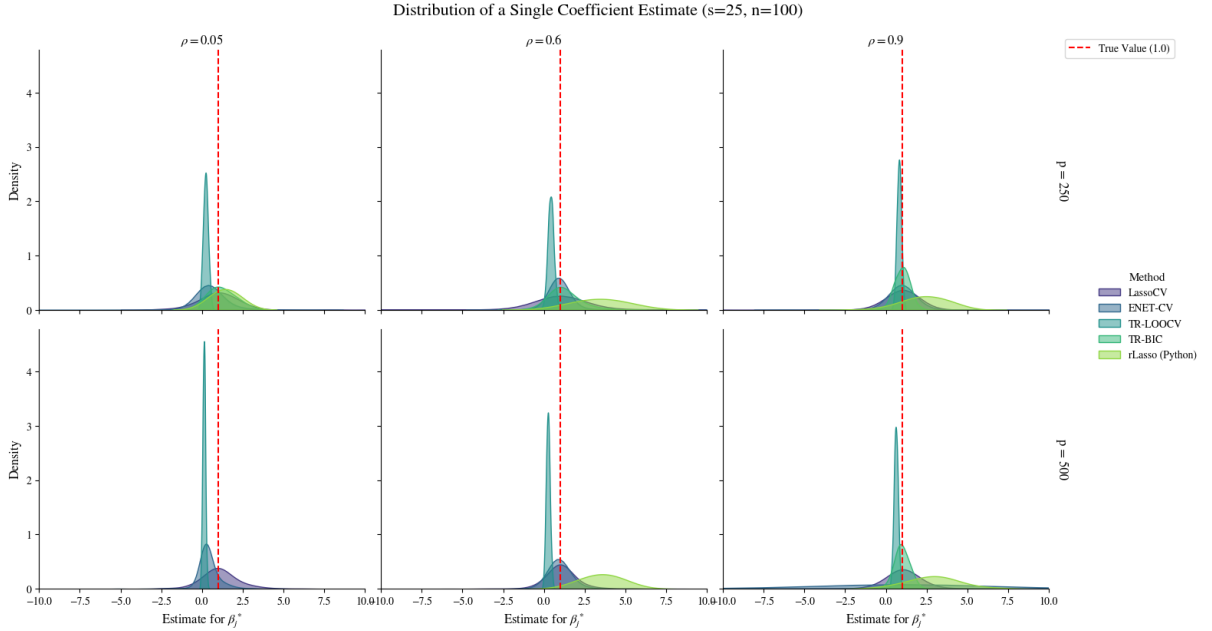


Figure 4: Sampling Distribution of a Single Coefficient Estimate for $\beta_j^* = 1.0$

5.3 Implications for Post-Selection Inference

The results show that selectors optimised for prediction are poor candidates for inference. The significant attenuation bias in the coefficient estimates from TR-LOOCV, visible in Figure 4, would lead to naive confidence intervals with severe under-coverage. Further-

more, the high model selection instability of the ENET-CV (as seen in Figure 1) would lead to poor inference.

In contrast, the TR-BIC estimator emerges as the most promising candidate for reliable subsequent inference under high-multicollinearity. By delivering a model that is both parsimonious (low FPR) and accurate (high TPR), it simultaneously mitigates the risk of omitted variable bias and the estimation instability. The resulting low-bias coefficient estimates seen in Figure 4 provide a sound basis for constructing valid confidence intervals. LassoCV also proves to be a robust contender. While it does not achieve the same degree of parsimony as TR-BIC, its coefficient estimates are similarly low-bias, and its performance is notably more stable across the different correlation structures examined. It therefore represents a strong benchmark.

While advanced techniques like the double-selection procedure (Belloni et al., 2014) are designed to be robust to minor selection errors, their finite-sample performance is nevertheless contingent on the quality of the first-stage selector. Our results suggest that employing TR-BIC as the selector within such frameworks could lead to more reliable and precise causal estimates in empirical applications characterized by high dimensionality and severe multicollinearity.

6 Empirical Application: Institutions

The simulation results from Section 5 show the superior model selection properties of the TR-BIC under multicollinearity. This section applies TR-BIC to a canonical problem in development economics and evaluates its performance within a post-double-selection framework.

6.1 Post-Selection Inference and the Double-Selection Framework

As discussed in Section 2.3, standard statistical inference following a data-driven model selection step is invalid.

To obtain valid inference on a single target parameter in a high-dimensional setting, we can use the double-selection procedure of Belloni, Chernozhukov, and Hansen, 2014.

The central hypothesis of this application is that TR-BIC, due to its superior support recovery under multicollinearity demonstrated in Section 5, can serve as a more effective selector than LASSO.

6.2 Empirical Application

We use the dataset from Acemoglu, Johnson, and Robinson (AJR, 2001) to re-examine the causal effect of institutions on economic growth. The model is specified as:

$$\text{logpgp95}_i = \alpha \cdot \text{avexpr}_i + W_i' \beta + \epsilon_i$$

The outcome, logpgp95_i , is the log of GDP per capita in 1995. The endogenous regressor, avexpr_i , is an index of institutional quality (protection against expropriation risk), which is instrumented by the log of settler mortality, logem4_i . The matrix W_i contains a set of 24 potential control variables. We have 64 observations (See Appendix D for details)

The primary challenge in this application, beyond endogeneity, is the degree of multicollinearity among the potential controls. Figure 5 displays the sample correlation matrix for controls. The heatmap shows strong positive correlation between temperature variables ('temp1'-'temp4') and among humidity variables ('humid1'-'humid4').

6.3 Methodology

To obtain a credible estimate of the causal effect of institutions on growth, we must simultaneously address the endogeneity of the institutional quality variable (avexpr) and the relatively *high*-dimensionality of the potential control variables (W_i). We implement the Post-Double-Selection procedure for Instrumental Variables (IV). This section first outlines the underlying theory and then details two alternative implementations.

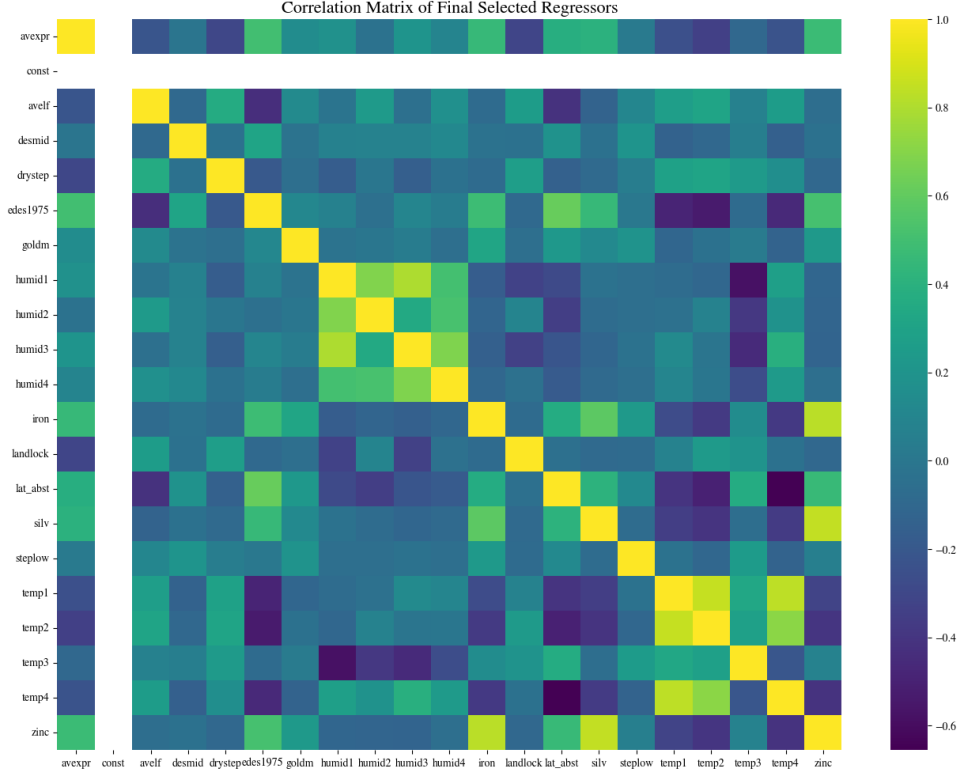


Figure 5: Correlation Matrix of covariates

Post-Double-Selection IV

We begin with the structural model for economic development where we are interested in the causal parameter α :

$$y_i = \alpha D_i + W_i' \beta + \epsilon_i$$

Here, y_i is the outcome (logpgp95), D_i is the endogenous treatment variable (avexpr), and W_i is a p -dimensional vector of potential control variables. To address the endogenous D_i , we use the log of settler mortality ($Z_i = \text{logem4}_i$) as an instrumental variable for institutional quality.

In a high-dimensional setting where the number of potential controls p is large relative to n , including all in a standard Two-Stage Least Squares (2SLS) regression risks overfitting. The Post-Double-Selection method provides a data-driven way to select the relevant controls for valid inference. The procedure is as follows:

1. The Two Selection Steps Instead of selecting controls based only on the structural equation, the method performs two separate variable selection steps using a high-dimensional selector (such as LASSO or, TR-BIC).

- **First Selection (Outcome Model):** We select a set of controls that are predictive of the outcome variable, y_i . We estimate a model of the form $y_i \sim W_i$ and identify

the set of indices for the selected controls:

$$\hat{S}_y = \{j \in \{1, \dots, p\} \mid \text{the } j\text{-th control is selected}\}$$

- **Second Selection (Treatment Model):** We repeat the process, this time selecting a set of controls that are predictive of the endogenous treatment variable, D_i . We estimate a model of the form $D_i \sim W_i$ and identify the corresponding set of indices:

$$\hat{S}_D = \{j \in \{1, \dots, p\} \mid \text{the } j\text{-th control is selected}\}$$

2. Union of Controls The final set of controls to be included in the IV regression is the union of the two selected sets. This is the crucial 'double-selection' step that ensures potential confounders are considered.

$$\hat{S}_{\text{final}} = \hat{S}_y \cup \hat{S}_D$$

Let $W_i[\hat{S}_{\text{final}}]$ denote the vector of control variables corresponding to this final selected set.

3. Final Post-Selection IV Estimation We then estimate the parameter α via a standard 2SLS procedure using only the selected controls, $W_i[\hat{S}_{\text{final}}]$.

- **First Stage (Post-Selection):** Regress the endogenous variable D_i on the instrument Z_i and the selected controls $W_i[\hat{S}_{\text{final}}]$:

$$D_i = Z_i\pi_Z + W_i[\hat{S}_{\text{final}}]'\pi_W + u_i$$

From this regression, we obtain the predicted values for the endogenous variable, $\hat{D}_i$.

- **Second Stage (Post-Selection):** Regress the outcome variable y_i on the predicted values $\hat{D}_i$ and the selected controls $W_i[\hat{S}_{\text{final}}]$:

$$y_i = \alpha\hat{D}_i + W_i[\hat{S}_{\text{final}}]'\beta_{\text{final}} + v_i$$

The resulting coefficient, $\hat{\alpha}$, is the Post-Double-Selection IV estimator of the causal effect. The intuition behind this procedure is that by selecting all controls related to *either* the outcome or the treatment, we make the final estimation of α robust to small errors in the selection process. This property, known as Neyman Orthogonality (Chernozhukov, 2018), ensures that the resulting estimator is approximately unbiased and allows for valid inference.

We implement and compare two versions of this procedure to choose the control variables from W_i :

1. **TR-BIC Selector:** Our proposed method, implemented in Python. We use the **Threshold Ridge with BIC** selector to perform the two selection steps. The final IV regression is estimated on the union of the selected controls using the IV2SLS package.
2. **LASSO Selector:** We use the standard `ivlasso` package in Stata as a benchmark. This package automates the Post-Double-Selection procedure for IV, using LASSO as the selector.

6.4 Empirical Results

Table 3: Comparative Results for the Causal Effect of Institutions on Income

Metric	TR-BIC Selector (DML)	LASSO Selector (ivlasso)	AJR (2001) Baseline
Point Estimate (α)	0.702	0.880	0.94
Standard Error	(0.163)	(0.273)	(0.16)
P-value	0.000	0.001	0.001
95% Confidence Interval	[0.382, 1.021]	[0.345, 1.414]	[0.626, 1.254]
No. of Controls Selected	19	5	0

The empirical results are summarised in Table 3. All methods confirm a positive and highly statistically significant effect of institutions on growth, consistent with the original findings of Acemoglu et al. (2001). The baseline 2SLS estimate in AJR (2001, Table 4) found a coefficient of 0.94 (s.e. 0.16). While our estimates are qualitatively similar, the choice of a high-dimensional selector yields slightly different conclusions.

The TR-BIC procedure yields a point estimate for the effect of institutions of **0.702** with a standard error of **0.163**. The method selects a union of 19 control variables, retaining correlated groups of geographic predictors, including four of five temperature variables and all four humidity variables. By conditioning on this comprehensive set of controls, the TR-BIC procedure may be argued as more effectively isolating variation, resulting in a more precise final estimate.

In contrast, the standard double-selection procedure using LASSO (`ivlasso`) produces a substantially larger point estimate of **0.880** with a much larger standard error of **0.273**. Consistent with the known instability of LASSO under multicollinearity, the `ivlasso` selector chooses a sparser set of only 5 controls, dropping many of the climate related variables retained by TR-BIC. The exclusion of these controls likely introduces some omitted variable bias, which can inflate both the point estimate and its standard error.

6.5 Discussion of Empirical Findings

The empirical application appears to confirm our simulation findings, suggesting that the TR-BIC procedure can yield a more precise causal estimate than the standard LASSO-based approach. A likely explanation for this difference in performance is how each method handles multicollinearity during the variable selection stage.

TR-BIC’s higher precision likely stems from its ability to retain a full group of correlated climate variables. This may lead to a more robust set of controls and, in turn, a more accurate final IV regression. LASSO, by contrast, is known to be unstable when variables are correlated and therefore tends to select a smaller, possibly incomplete, set of controls. The exclusion of these potentially important variables likely introduces omitted variable bias, which would serve to increase both the final point estimate and its standard error.

In addition, the difference in model size, 19 controls for TR-BIC versus 5 for LASSO, calls into question the validity of the sparsity assumption. LASSO is built on the premise that the true model is sparse. However, TR-BIC, which our simulations suggest has superior support recovery capabilities, selects a much denser model. This finding indicates that the true data generating process for economic growth might not be sparse. This is a plausible outcome given that many factors are likely predictive of growth (Xavier Sala-i-Martin, 1997).

This result highlights the need for variable selectors that are robust to both multicollinearity and potential violations of the sparsity assumption.

7 Conclusion and Future Research

This dissertation introduces the Threshold Ridge estimator tuned via the Bayesian Information Criterion (TR-BIC) as a variable selection method for high-dimensional linear models under multicollinearity. Simulations show that under severe multicollinearity, TR-BIC achieves superior support recovery and yields more precise coefficient estimates than LASSO and TR-LOOCV. In low-collinearity settings, however, TR-BIC can be prone to under-selection. An empirical application to the Acemoglu et al. 2001 dataset confirms these findings; the TR-BIC selector produces a more precise estimate of the institutional effect on growth by better handling multicollinearity among geographic controls. It also challenges the validity of the underlying sparsity assumption by selecting a denser model.

Future work should focus on several areas. First, a formal theoretical analysis is required to establish the asymptotic properties and convergence rates of the TR-BIC estimator. Building on this, research could develop tailored post-selection inference methods for Threshold Ridge estimators, adapting existing conditional or debiased frameworks. Finally, more extensive empirical testing is needed to characterise the performance of

TR-BIC under alternative data structures, such as those with heavy-tailed distributions or heteroskedastic errors.

A Algorithms for TR Tuning

Algorithm 1 Threshold Ridge with LOOCV Tuning (TR-LOOCV)

- 1: **Input:** Data (X, y) , grid of ridge penalties $H = \{h_1, \dots, h_M\}$, grid of thresholds $A = \{a_1, \dots, a_K\}$.
 - 2: **Initialize:** $CV_{min} \leftarrow \infty$, $\beta_{best} \leftarrow \mathbf{0}$, $h_{best} \leftarrow h_1$, $a_{best} \leftarrow a_1$.
 - 3: **Initialize dictionaries:** $\Theta_{ridge} \leftarrow \{\}$, $W_{diag} \leftarrow \{\}$.
 - 4: {Pre-compute Ridge fits and hat matrix diagonals for each h }
 - 5: **for** $h \in H$ **do**
 - 6: Compute Ridge coefficients: $\hat{\theta}(h) = (X^\top X + hI_p)^{-1} X^\top y$.
 - 7: Store $\Theta_{ridge}[h] \leftarrow \hat{\theta}(h)$.
 - 8: Compute hat matrix diagonal: $w(h) = \text{diag}(X(X^\top X + hI_p)^{-1} X^\top)$.
 - 9: Store $W_{diag}[h] \leftarrow w(h)$.
 - 10: **end for**
 - 11: {Grid search for optimal (h, a) based on CV score}
 - 12: **for** $h \in H$ **do**
 - 13: **for** $a \in A$ **do**
 - 14: Get pre-computed $\hat{\theta}(h)$ from Θ_{ridge} and $w(h)$ from W_{diag} .
 - 15: Apply hard thresholding: $\tilde{\beta}_j(h, a) \leftarrow \hat{\theta}_j(h) \cdot \mathbf{1}\{|\hat{\theta}_j(h)| > a\}$ for $j = 1, \dots, p$.
 - 16: Calculate residuals: $e = y - X\tilde{\beta}(h, a)$.
 - 17: Calculate LOOCV score proxy: $CV(h, a) = \frac{1}{n} \sum_{i=1}^n \left(\frac{e_i}{1-w_i(h)} \right)^2$.
 - 18: **if** $CV(h, a) < CV_{min}$ **then**
 - 19: $CV_{min} \leftarrow CV(h, a)$.
 - 20: $\beta_{best} \leftarrow \tilde{\beta}(h, a)$.
 - 21: $h_{best} \leftarrow h$.
 - 22: $a_{best} \leftarrow a$.
 - 23: **end if**
 - 24: **end for**
 - 25: **end for**
 - 26: **Output:** Optimal coefficients β_{best} , optimal ridge penalty h_{best} , optimal threshold a_{best} .
-

Algorithm 2 Threshold Ridge with BIC Tuning (TR-BIC)

```

1: Input: Data  $(X, y)$ , grid of ridge penalties  $H = \{h_1, \dots, h_M\}$ , grid of thresholds
    $A = \{a_1, \dots, a_K\}$ .
2: Initialize:  $BIC_{min} \leftarrow \infty$ ,  $\beta_{best} \leftarrow \mathbf{0}$ ,  $h_{best} \leftarrow h_1$ ,  $a_{best} \leftarrow a_1$ .
3: Initialize dictionary:  $\Theta_{ridge} \leftarrow \{\}$ .
4: {Pre-compute Ridge fits for each  $h$ }
5: for  $h \in H$  do
6:   Compute Ridge coefficients:  $\hat{\theta}(h) = (X^\top X + hI_p)^{-1} X^\top y$ .
7:   Store  $\Theta_{ridge}[h] \leftarrow \hat{\theta}(h)$ .
8: end for
9: {Grid search for optimal  $(h, a)$  based on BIC score}
10: for  $h \in H$  do
11:   for  $a \in A$  do
12:     Get pre-computed  $\hat{\theta}(h)$  from  $\Theta_{ridge}$ .
13:     Apply hard thresholding:  $\tilde{\beta}_j(h, a) \leftarrow \hat{\theta}_j(h) \cdot \mathbf{1}\{|\hat{\theta}_j(h)| > a\}$  for  $j = 1, \dots, p$ .
14:     Calculate Residual Sum of Squares:  $RSS = \|y - X\tilde{\beta}(h, a)\|_2^2$ .
15:     Calculate model degrees of freedom:  $k = \|\tilde{\beta}(h, a)\|_0 + 1$ .
16:     Calculate BIC score:  $BIC(h, a) = n \cdot \ln(RSS/n) + k \cdot \ln(n)$ .
17:     if  $BIC(h, a) < BIC_{min}$  then
18:        $BIC_{min} \leftarrow BIC(h, a)$ .
19:        $\beta_{best} \leftarrow \tilde{\beta}(h, a)$ .
20:        $h_{best} \leftarrow h$ .
21:        $a_{best} \leftarrow a$ .
22:     end if
23:   end for
24: end for
25: Output: Optimal coefficients  $\beta_{best}$ , optimal ridge penalty  $h_{best}$ , optimal threshold
    $a_{best}$ .

```

B Derivation of OLS Risk Bound

We consider a linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is an $n \times p$ matrix, $\boldsymbol{\varepsilon} \sim N(0, \sigma^2 \mathbf{I}_n)$, and the true coefficient vector $\boldsymbol{\beta}^*$ is sparse with support $S^* = \{j : \beta_j^* \neq 0\}$ and size $|S^*| = s$.

The Oracle OLS estimator, which knows the true support S^* , is defined for the non-zero coefficients as:

$$\hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} = (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \mathbf{y} \quad (1)$$

For all indices $j \notin S^*$, the oracle estimate is zero, $\hat{\beta}_{j, \text{oracle}} = 0$. Since the true coefficients are also zero for $j \notin S^*$, the model can be written using only the s relevant predictors:

$$\mathbf{y} = \mathbf{X}_{S^*} \boldsymbol{\beta}_{S^*}^* + \boldsymbol{\varepsilon} \quad (2)$$

To find the estimation error, we substitute the model into the estimator equation:

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} &= (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top (\mathbf{X}_{S^*} \boldsymbol{\beta}_{S^*}^* + \boldsymbol{\varepsilon}) \\ &= (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \mathbf{X}_{S^*} \boldsymbol{\beta}_{S^*}^* + (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \boldsymbol{\varepsilon} \\ &= \mathbf{I}_s \boldsymbol{\beta}_{S^*}^* + (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \boldsymbol{\varepsilon} \\ &= \boldsymbol{\beta}_{S^*}^* + (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \boldsymbol{\varepsilon} \end{aligned}$$

The estimation error for the non-zero coefficients is therefore:

$$\hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} - \boldsymbol{\beta}_{S^*}^* = (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \boldsymbol{\varepsilon} \quad (3)$$

For the full p -dimensional vector, the error $\hat{\boldsymbol{\beta}}_{\text{oracle}} - \boldsymbol{\beta}^*$ has non-zero entries only at indices in S^* .

We now evaluate the in-sample prediction risk, defined as the expected squared prediction error averaged over the training samples. This risk measures how well the model fits the data it was trained on.

$$\text{Risk} = \frac{1}{n} E \left[\|\mathbf{X}(\hat{\boldsymbol{\beta}}_{\text{oracle}} - \boldsymbol{\beta}^*)\|_2^2 \right] \quad (4)$$

Since the estimation error is zero outside the true support S^* , we can write $\mathbf{X}(\hat{\boldsymbol{\beta}}_{\text{oracle}} - \boldsymbol{\beta}^*) = \mathbf{X}_{S^*}(\hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} - \boldsymbol{\beta}_{S^*}^*)$. Substituting this into the risk gives:

$$\text{Risk} = \frac{1}{n} E \left[\|\mathbf{X}_{S^*}(\hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} - \boldsymbol{\beta}_{S^*}^*)\|_2^2 \right] \quad (5)$$

To simplify this expression, we introduce the hat matrix $\mathbf{H}_{S^*} = \mathbf{X}_{S^*}(\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top$, which projects onto the column space of $\mathbf{X}_{S^*}$. The term inside the norm becomes:

$$\mathbf{X}_{S^*}(\hat{\boldsymbol{\beta}}_{S^*, \text{oracle}} - \boldsymbol{\beta}_{S^*}^*) = \mathbf{X}_{S^*} ((\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \boldsymbol{\varepsilon}) = \mathbf{H}_{S^*} \boldsymbol{\varepsilon} \quad (6)$$

The squared norm is therefore $\|\mathbf{H}_{S^*}\boldsymbol{\varepsilon}\|_2^2 = (\mathbf{H}_{S^*}\boldsymbol{\varepsilon})^\top (\mathbf{H}_{S^*}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}^\top \mathbf{H}_{S^*}^\top \mathbf{H}_{S^*} \boldsymbol{\varepsilon}$. Since $\mathbf{H}_{S^*}$ is a projection matrix, it is symmetric ($\mathbf{H}_{S^*}^\top = \mathbf{H}_{S^*}$) and idempotent ($\mathbf{H}_{S^*}^2 = \mathbf{H}_{S^*}$), so the expression simplifies to:

$$\|\mathbf{H}_{S^*}\boldsymbol{\varepsilon}\|_2^2 = \boldsymbol{\varepsilon}^\top \mathbf{H}_{S^*} \boldsymbol{\varepsilon} \quad (7)$$

The risk is now expressed as the expectation of this quadratic form:

$$\text{Risk} = \frac{1}{n} E [\boldsymbol{\varepsilon}^\top \mathbf{H}_{S^*} \boldsymbol{\varepsilon}] \quad (8)$$

We evaluate the expectation using the trace trick for a random vector $\mathbf{z}$ and a constant matrix $\mathbf{A}$: $E[\mathbf{z}^\top \mathbf{A} \mathbf{z}] = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{z})) + E[\mathbf{z}]^\top \mathbf{A} E[\mathbf{z}]$. Here, $\mathbf{z} = \boldsymbol{\varepsilon}$ and $\mathbf{A} = \mathbf{H}_{S^*}$. With our assumptions $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$, we get:

$$\begin{aligned} E [\boldsymbol{\varepsilon}^\top \mathbf{H}_{S^*} \boldsymbol{\varepsilon}] &= \text{tr}(\mathbf{H}_{S^*} \text{Cov}(\boldsymbol{\varepsilon})) + \mathbf{0}^\top \mathbf{H}_{S^*} \mathbf{0} \\ &= \text{tr}(\mathbf{H}_{S^*} \sigma^2 \mathbf{I}_n) \\ &= \sigma^2 \text{tr}(\mathbf{H}_{S^*}) \end{aligned}$$

The trace of the hat matrix is equal to the dimension of the subspace it projects onto, which is the number of columns in $\mathbf{X}_{S^*}$:

$$\begin{aligned} \text{tr}(\mathbf{H}_{S^*}) &= \text{tr}(\mathbf{X}_{S^*} (\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top) \\ &= \text{tr}((\mathbf{X}_{S^*}^\top \mathbf{X}_{S^*})^{-1} \mathbf{X}_{S^*}^\top \mathbf{X}_{S^*}) \quad (\text{using } \text{tr}(ABC) = \text{tr}(CAB)) \\ &= \text{tr}(\mathbf{I}_s) = s \end{aligned}$$

Finally, substituting these results back into the expression for risk yields the bound:

$$\text{Risk} = \frac{1}{n} (\sigma^2 \text{tr}(\mathbf{H}_{S^*})) = \frac{\sigma^2 s}{n} \quad (9)$$

This result shows that the in-sample prediction risk of the oracle estimator is directly proportional to the error variance and the model sparsity, and inversely proportional to the sample size.

C Additional Simulation Results and Plots

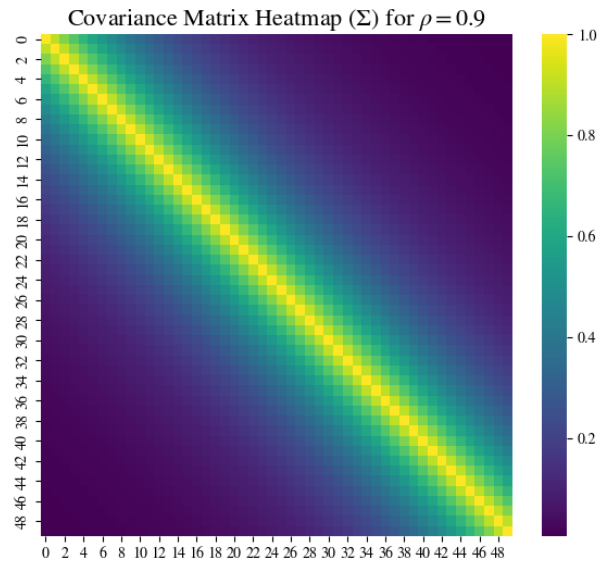


Figure 6: High Multicollinearity Matrix $\rho = 0.9$

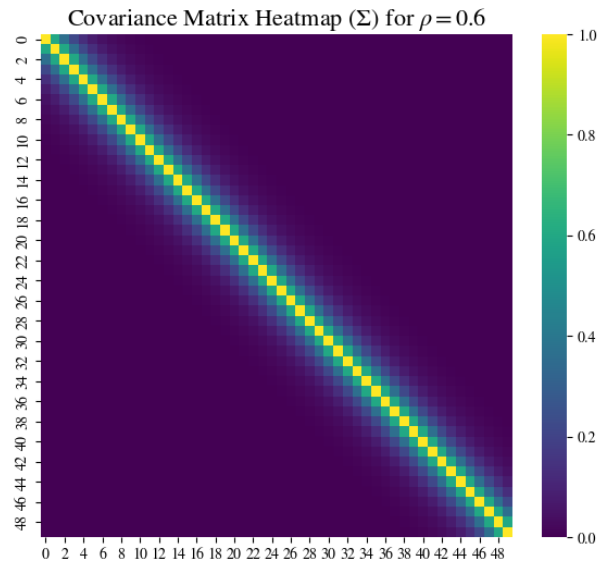


Figure 7: Medium Multicollinearity Matrix $\rho = 0.6$

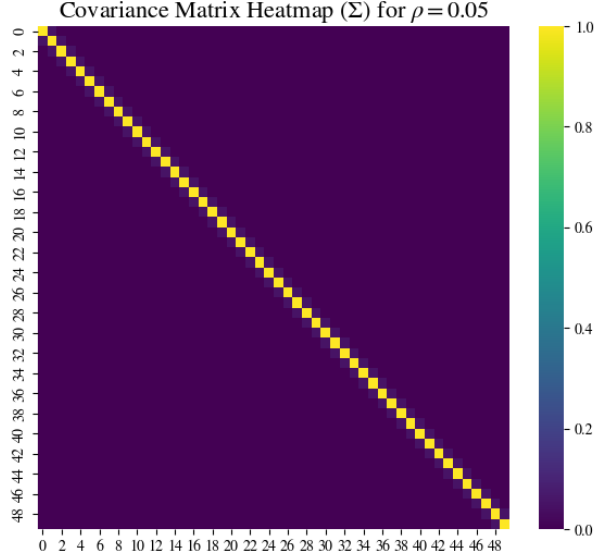


Figure 8: Low Multicollinearity Matrix $\rho = 0.05$

Table 4: Aggregated Simulation Results by Scenario

Method	p	ρ	n	s	TPR	FPR	Model Size	Pred. Risk	Oracle Risk	Est. Error
<i>Scenario: $p = 250$</i>										
ENET-CV	250	0.05	100	25	0.672	0.368	99.63	78.69	1.05	80.10
LassoCV	250	0.05	100	25	0.324	0.070	23.86	85.70	1.05	87.00
TR-BIC	250	0.05	100	25	0.025	0.000	0.71	27.19	1.05	24.81
TR-LOOCV	250	0.05	100	25	0.979	0.930	233.82	21.29	1.05	19.78
rLasso	250	0.05	100	25	0.002	0.000	0.05	27.49	1.05	24.99
ENET-CV	250	0.60	100	25	0.954	0.162	60.24	58.92	1.02	90.29
LassoCV	250	0.60	100	25	0.886	0.107	46.30	1014.17	1.02	1552.45
TR-BIC	250	0.60	100	25	0.294	0.000	7.44	48.19	1.02	32.45
TR-LOOCV	250	0.60	100	25	1.000	0.950	238.78	28.13	1.02	11.84
rLasso	250	0.60	100	25	0.041	0.000	1.03	80.52	1.02	31.65
ENET-CV	250	0.90	100	25	0.995	0.079	42.61	11.44	1.04	70.00
LassoCV	250	0.90	100	25	0.909	0.055	35.11	26.84	1.04	160.35
TR-BIC	250	0.90	100	25	1.000	0.007	26.67	1.23	1.04	11.09
TR-LOOCV	250	0.90	100	25	1.000	0.928	233.76	5.82	1.04	3.76
rLasso	250	0.90	100	25	0.278	0.000	6.97	23.43	1.04	85.60
<i>Scenario: $p = 500$</i>										
ENET-CV	500	0.05	100	25	0.507	0.227	120.29	549.95	1.06	611.96
LassoCV	500	0.05	100	25	0.181	0.028	17.82	63.09	1.06	62.00
TR-BIC	500	0.05	100	25	0.000	0.000	0.01	27.59	1.06	25.00
TR-LOOCV	500	0.05	100	25	0.954	0.866	435.28	24.32	1.06	22.29
rLasso	500	0.05	100	25	0.001	0.000	0.03	27.59	1.06	25.00
ENET-CV	500	0.60	100	25	0.880	0.086	63.06	144.89	1.04	251.25
LassoCV	500	0.60	100	25	0.796	0.057	46.76	131.55	1.04	219.46
TR-BIC	500	0.60	100	25	0.010	0.000	0.24	89.89	1.04	26.84
TR-LOOCV	500	0.60	100	25	0.999	0.920	462.19	50.44	1.04	16.87
rLasso	500	0.60	100	25	0.028	0.000	0.70	84.38	1.04	30.05
ENET-CV	500	0.90	100	25	0.993	0.049	48.32	1412.04	1.04	10015.97
LassoCV	500	0.90	100	25	0.896	0.033	38.19	56.15	1.04	272.79
TR-BIC	500	0.90	100	25	0.964	0.002	24.85	4.06	1.04	15.24
TR-LOOCV	500	0.90	100	25	1.000	0.920	461.82	31.68	1.04	7.16
rLasso	500	0.90	100	25	0.246	0.000	6.17	28.94	1.04	93.66

D Empirical Application Details

Table 5: AJR (2001) Variable Descriptions

Variable	Description
<i>Panel A: Variables of Interest</i>	
logpgp95	Log GDP per capita in 1995.
avexpr	Average protection against expropriation risk, 1985-95.
logem4	Log of settler mortality.
<i>Panel B: Geographic and Climate Controls</i>	
lat_abst	Absolute latitude, scaled from 0 to 1.
temp1 - temp5	Set of five temperature indicators.
humid1 - humid4	Set of four humidity indicators.
steplow - drywint	Set of six soil type and climate indicators.
landlock	Dummy variable equal to 1 if country is landlocked.
goldm - oilres	Set of five indicators for mineral and natural resource wealth.
<i>Panel C: Historical and Political Controls</i>	
edes1975	Percent of population of European descent in 1975.
avelf	Ethno-linguistic fractionalization index.
indtime	Years of independence as of 1995.
euro1900	Percent of population who were European settlers in 1900.
democ1	Democracy index in first year of independence.
cons1	Constraint on executive index in first year of independence.
democ00a	Democracy index in 1900.
cons00a	Constraint on executive index in 1900.

Table 6: Control Variables Selected by TR-BIC and LASSO

TR-BIC	LASSO
avelf	avelf
desmid	edes1975
drystep	iron
edes1975	temp2
goldm	zinc
humid1	
humid2	
humid3	
humid4	
iron	
landlock	
lat_abst	
silv	
steplow	
temp1	
temp2	
temp3	
temp4	
zinc	
Total Controls: 19	Total Controls: 5